

**BACHELOR OF COMPUTER
APPLICATIONS (BCA) (REVISED)**

Term-End Examination

June, 2023

**BCS-051 : INTRODUCTION TO SOFTWARE
ENGINEERING**

Time : 3 Hours

Maximum Marks : 100

Weightage : 75%

Note : *Question No. 1 is compulsory and carries
40 marks. Attempt any **three** questions from
the rest.*

1. (a) Develop SRS for Online Study Center Allocation System (OSCAS) for students who apply for admission to a university. SRS should be in IEEE format. Make necessary assumptions. 25

- (b) Explain Iterative Enhancement Model with the help of an example. What are its advantages and disadvantages over Spiral Model ? 10
- (c) List the diagrams whose specifications are supported by UML. 5
- 2. (a) Draw the zero and first level DFDs for OSCAS. Make necessary assumptions. 10
- (b) Draw PERT chart for the development of OSCAS. 10
- 3. (a) Draw ERD for OSCAS. Make necessary assumptions. 10
- (b) Draw usecase diagram for OSCAS. 10
- 4. (a) Explain the five solution design principles in function oriented design. 10
- (b) Explain the specification of a static object. 10
- 5. (a) Explain the *five* maturity levels of SEI-CMM. 10
- (b) Briefly, explain ‘Step-Wise’ framework. 10

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Maximum Marks : 100

Weightage : 75%

Note : *Question No. 1 is compulsory and carries 40 marks. Attempt any **three** questions from the rest.*

1. (a) Develop SRS for *Online Admission System (OAS)* for a university. SRS should be in IEEE format. Make necessary assumptions. 25
- (b) Explain *Spiral Model* with the help of an example. What are its advantages and disadvantages over *Waterfall Model* ? 10

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- (c) What is a Data Dictionary ? What are its contents ? 5
2. (a) Draw the zero and first level DFDs for OAS. Make necessary assumptions. 10
- (b) Draw PERT chart for the development of OAS. 10
3. (a) Draw ERD for OAS. Make necessary assumptions. 10
- (b) Draw use case diagram for OAS. 10
4. (a) What is meant by function oriented design ? Explain the concepts associated with function oriented design. 10
- (b) What is meant by static object ? How will you identify them ? Explain. 10
5. (a) Explain the terms 'Alpha testing' and 'Beta testing'. 10
- (b) What is meant by cohesion ? Explain different types of cohesion. 10

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1. (a) Develop SRS for *Online Assignment Submission System (OASS)* for students of various courses of university. SRS should be in IEEE format. Make necessary assumptions. 25

- (b) Explain Rapid Application Development (RAD) approach for Software Development. What are the advantages and disadvantages of RAD approach over Waterfall model. 10
- (c) Explain any *five* key features of Function Oriented Design. 5
- 2. (a) Draw the zero and first level DFDs for OASS. Make necessary assumptions. 10
- (b) Draw GANTT chart for the development of OASS. 10
- 3. (a) Draw ERD for OASS. Make necessary assumptions. 10
- (b) Define *Data Dictionary*. Consider a Data Element *Student*. What information about *Student* is stored in Data Dictionary ? 10
- 4. (a) Explain the methodology for identification of objects in Object Oriented Design with an example. 10
- (b) Explain the terms 'Module Testing' and 'Integration Testing'. 10

5. (a) Explain various challenges for successful debugging. 10
- (b) List any *five* software testing tools indicating the testing phase in which they can be employed. 10

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1. (a) Explain spiral model for software development. Explain the types of software systems developed using this model. 10
- (b) What is PERT chart ? Explain the creation process of PERT chart, with suitable example. 10

- (c) Explain the terms coupling and cohesion, with the help of suitable example for each. Briefly discuss the impact of coupling and cohesion, on software development. 10
- (d) Draw the first two levels of DFDs for an “Online Student Registration System”. Make necessary assumptions required. 10
2. (a) What is a data dictionary ? Explain the contents of data dictionary with the help of an example. 10
- (b) What is module testing ? How is it different from integration testing ? Give suitable test case example for both types of testing. 10
3. (a) Develop the SRS for an “Online Student Registration System”. Make necessary assumptions. Follow IEEE SRS format. Discuss the characteristics of good SRS. 15
- (b) Discuss the application of logic objects, with the help of an example. 5

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4. (a) Define the term 'Software Quality'. Explain the McCall's software quality factors. 10
- (b) What is debugging ? Explain any *four* debugging strategies. 10
5. Write short notes on the following : 4×5=20
- (a) Capability Maturity Models (CMM)
- (b) Software Metrics
- (c) Entity Relationship Diagram
- (d) Class Diagram

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Time : 3 Hours

Maximum Marks : 100

Note : *Question No. 1 is compulsory and carries
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from the rest.*

1. (a) Compare Waterfall model and Spiral
model for software development.

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Explain the situation, in which the Spiral model for software development should be preferred over Waterfall model. 10

- (b) What is GANTT chart ? Compare the utility of GANTT and PERT charts. Draw GANTT chart for the development of "Online Railway Reservation System". 10
- (c) Discuss IEEE SRS format and use it to develop the SRS for "Online Railway Reservation System". Make necessary assumptions. 15
- (d) What are static objects ? Explain with the help of an suitable example. 5

2. (a) Briefly discuss the various notations used to draw usecase diagram. Draw a usecase diagram for “Online Railway Reservation System”. 10
- (b) Compare Black Box Testing and White Box Testing. Give suitable example for each. 10
3. (a) Briefly discuss the various notations of DFD. Draw first two levels of DFDs for an “Online Examination System”. Make necessary assumptions. 10
- (b) What is the need of project planning ? Discuss any *four* types of project plans.

10

4. (a) What is the need of software maintenance ? Explain different categories of software maintenance. 10
- (b) What is a test case ? Explain test case for functional testing with the help of an example. 10
5. Briefly discuss the following with suitable example for each : $4 \times 5 = 20$
- (a) Acceptance testing
- (b) Alpha and Beta testing
- (c) Module testing
- (d) Integration testing
- (e) Software Quality Assurance activities

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1. (a) Develop SRS for Online Assignment Submission and Evaluation System for a University. Use IEEE format for SRS. Make necessary assumptions. 20
- (b) What is an object ? Explain application logic object with an example. Also, explain specifications of static object. 10
- (c) What is structured analysis ? Explain. 5
- (d) Explain the principles of software testing. 5
2. (a) Explain Waterfall Model with the help of a suitable diagram. List advantages and disadvantages of this model. 10

- (b) Draw a sample Gantt chart for an online examination system. Make necessary assumptions. 10
3. (a) Explain System testing and Regression testing. 10
- (b) What is the need of software maintenance ? Briefly explain different types of software maintenance. 10
4. (a) What is Software Configuration Management (SCM) ? Explain SCM activities briefly. 10
- (b) What is a DFD ? What are the components of DFD ? Draw first two levels of DFDs for Online Assignment Submission and Evaluation System. Make necessary assumptions. 10

5. Write short notes on the following : $4 \times 5 = 20$

- (a) Reverse Engineering
- (b) Capability Maturity Model (CMM)
- (c) Debugging
- (d) Use-case diagram

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**BACHELOR OF COMPUTER
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Term-End Examination

June, 2026

**BCS-051 : INTRODUCTION TO SOFTWARE
ENGINEERING**

Time : 3 Hours

Maximum Marks : 100

Weightage : 75%

Note : Question No. 1 is compulsory and carries

*40 marks. Attempt any **three** questions*

from the rest.

1. (a) Develop SRS for online examination system for a university. Use IEEE format for SRS. Make necessary assumptions. 20
- (b) Draw ERD for online shopping system. Make necessary assumptions. 10
- (c) What is coupling ? How is it related to cohesion ? Explain. 5
- (d) Explain class diagram with the help of an example. 5
2. (a) What is a DFD ? What are the components of DFD ? Draw first two levels of DFA for online examination system. Make necessary assumptions. 10

- (b) What is Software Testing ? List the scope of Software Testing. Also, explain principles of Software Testing. 10
3. (a) Explain spiral model for software development. Also, describe the kind of system which should be developed using the model. List limitations of this model. 10
- (b) Explain software metrics in detail. 10
4. (a) Draw PERT chart/diagram for online examination system. Make necessary assumptions. 10
- (b) What is software quality assurance ? Explain the terms Error, Fault, Failure. What is cost of quality ? 10

5. Write short notes on the following : 4×5=20

- (a) Integration Testing
- (b) Object Oriented Design
- (c) Software Maintenance
- (d) Designing Test Cases

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